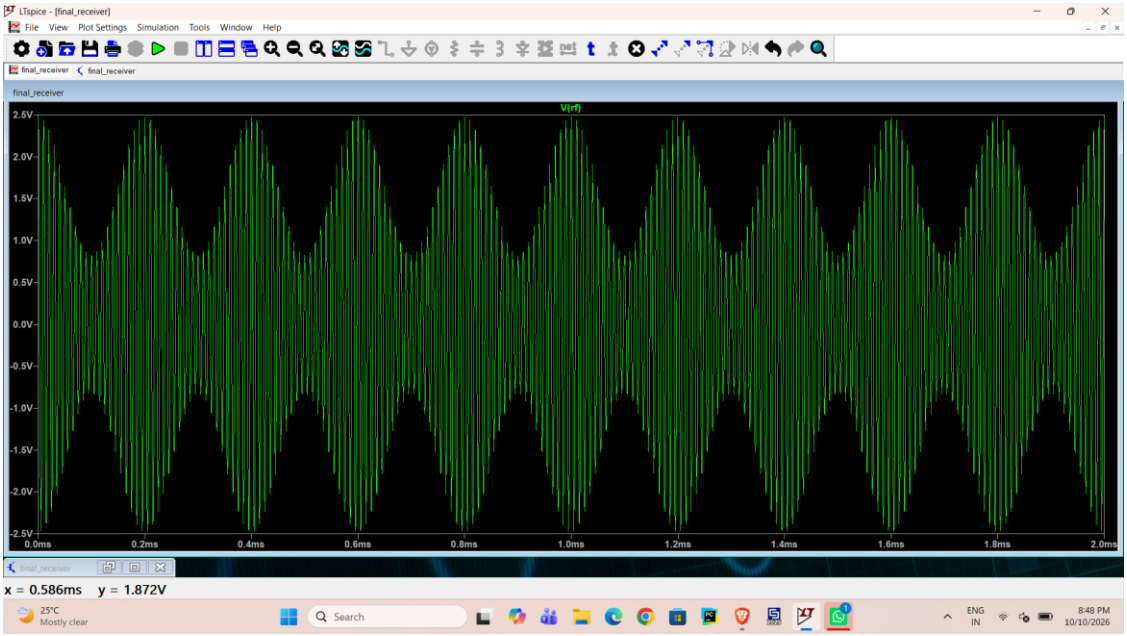
02. Results of Each Stage

All plots below are LTspice results from the final design. Numbers marked *ngspice* were checked with an independent ngspice run of the same transistor-level circuit. Numbers marked *plot* are read from the LTspice screenshot and are approximate. The schematics are in 03_circuits_and_design.docx.

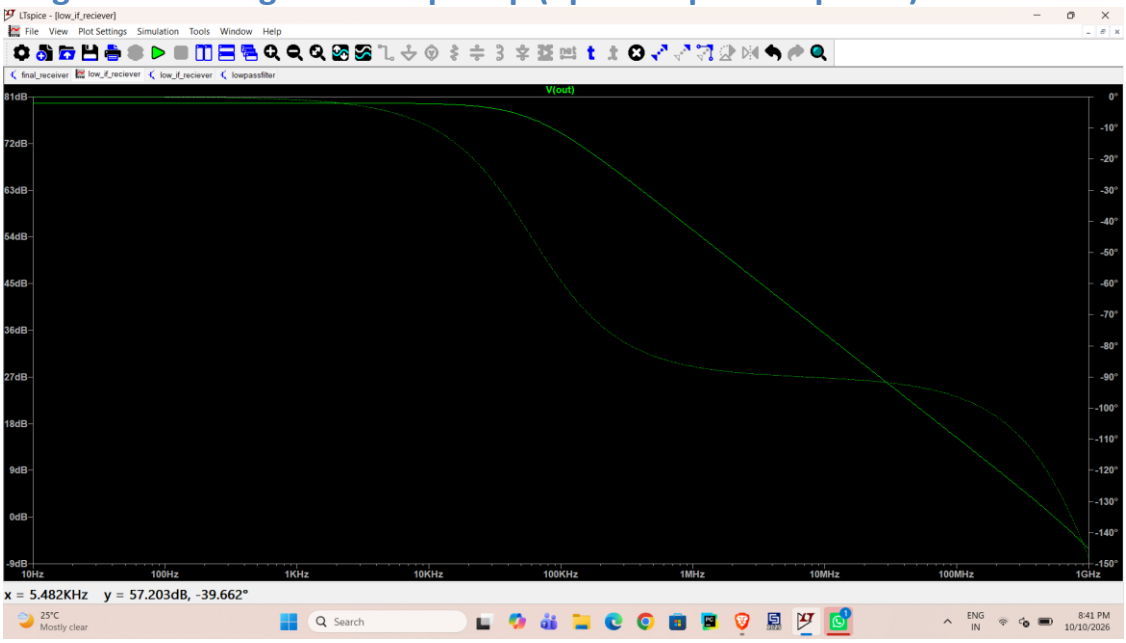
Stage 0: RF input signal (AM)



RF AM signal

What it shows - The 100 kHz carrier is amplitude-modulated by the 5 kHz message: the envelope repeats every 0.2 ms. - The envelope peaks at about 2.475 V and its valleys are at about 0.825 V (*plot*), which gives a modulation index μ = (2.475 − 0.825)/(2.475 + 0.825) = 0.5. - The signal is centered on 0 V, as an antenna signal would be. The mixers add the DC offsets later. - This is the input of the receiver. The goal is to recover the 5 kHz envelope information from it.

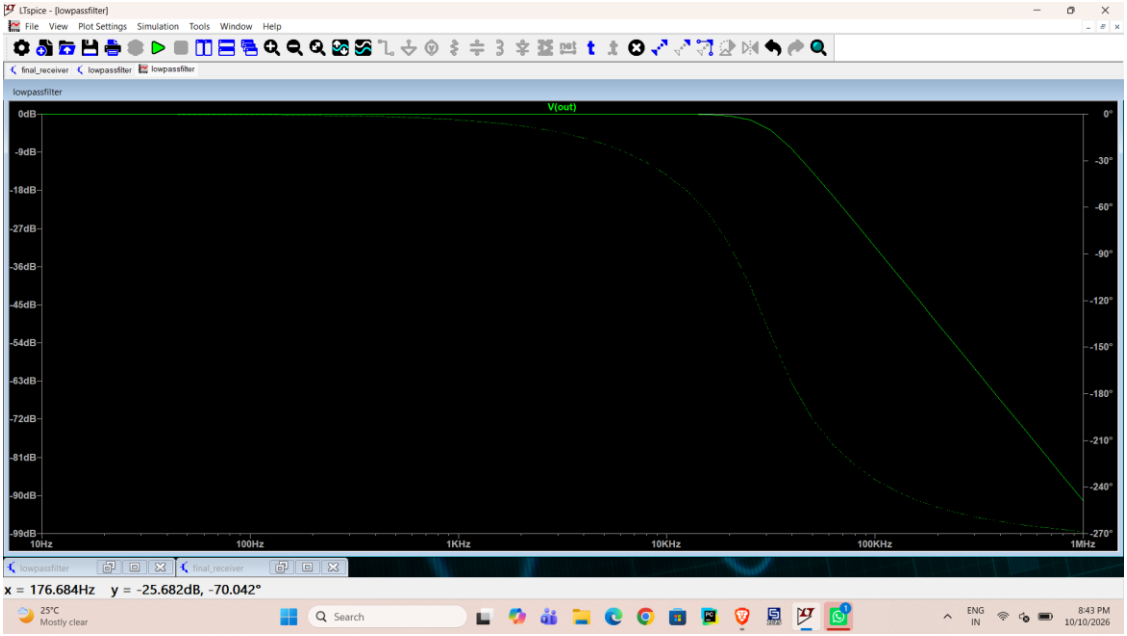
Stage 1: Two-stage CMOS op-amp (open-loop AC response)



Op-amp AC response

What it shows - The gain is flat at about **80 dB** at low frequency, then falls at 20 dB per decade after a dominant pole set by the Miller capacitor. *ngspice*: DC gain 9719 V/V (79.7 dB). - The phase falls from 0 degrees towards -90 degrees (dominant pole) and drops further at high frequency where the second pole acts. *ngspice* with Cc = 0.1 pF: unity-gain frequency about 206 MHz, with a unity-gain phase margin of about 13 degrees. - The op-amp draws 405 µA from 3.3 V and holds its output at 1.65 V when connected as a buffer (*ngspice*). - This high gain and wide gain-bandwidth let the active-RC filter follow its ideal response. Phase margin as a stand-alone amplifier is low, but inside the filter the feedback factor is below 1 and the results show no ringing.

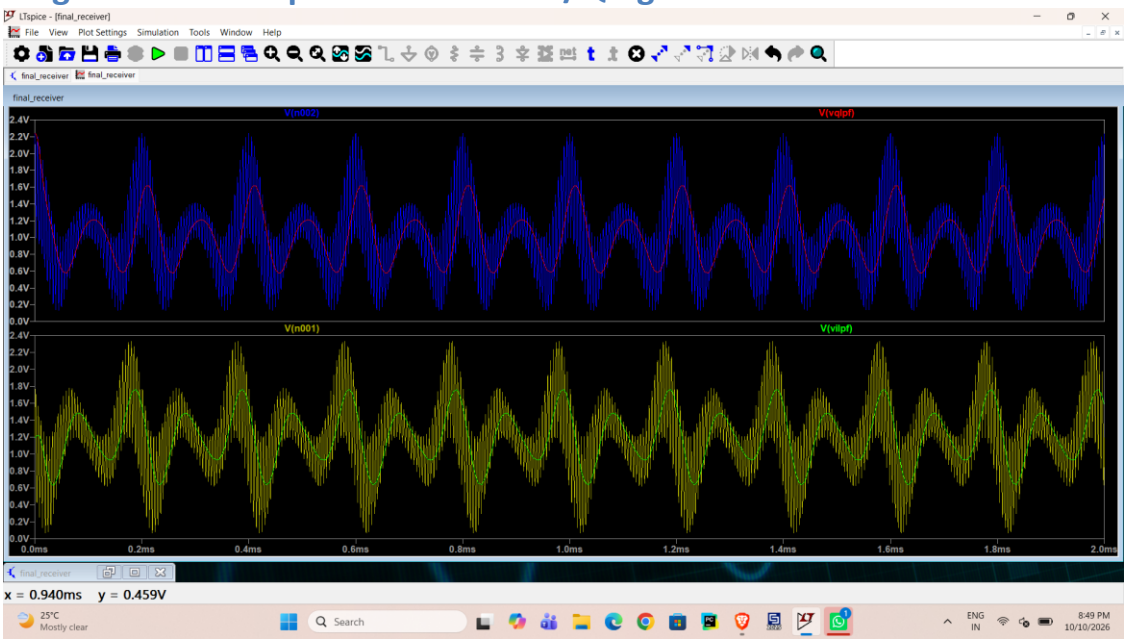
Stage 2: 3rd-order Butterworth low-pass filter (AC response)



Low-pass filter AC response

What it shows - The gain is flat at 0 dB in the passband and the **-3 dB frequency is 30 kHz** (*ngspice*: 29.9 kHz). - Gain at the frequencies of interest (*ngspice*): 10 kHz = 0.995, 15 kHz = 0.988 (message sidebands pass with about 0.1 dB loss), 100 kHz = 0.027 (-31 dB), 190 kHz = 0.004 (-48 dB). - The roll-off is 60 dB per decade, reaching about -91 dB at 1 MHz (*plot*), which is the 3rd-order slope. The phase goes towards -270 degrees (three poles), with about -39 degrees at the 10 kHz IF. - The 190 kHz mixer sum product is removed, while the 10 kHz IF and its 5 kHz sidebands pass.

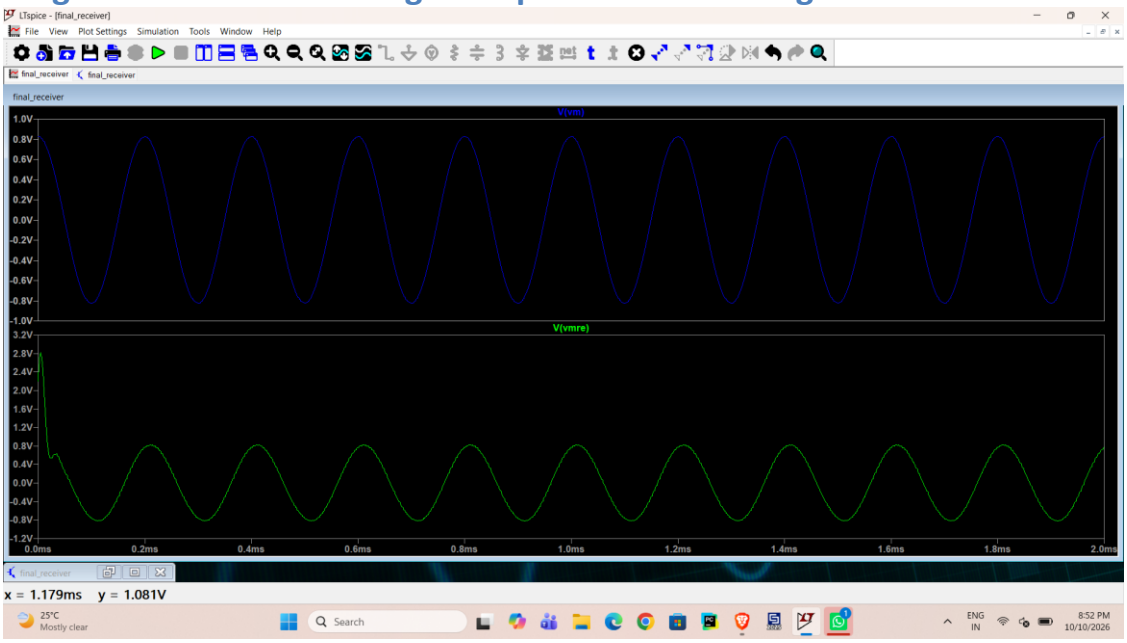
Stage 3: Mixer outputs and filtered I/Q signals



Mixer outputs and filtered I/Q

What it shows - Top panel is the Q path and bottom panel is the I path. The dense traces (`V(n002)`, `V(n001)`) are the mixer outputs before filtering: they swing about 0.1 V to 2.35 V and carry fast 190 kHz ripple on top of the wanted signal. - The smooth traces (`V(vqlpf)`, `V(vilpf)`) are the low-pass outputs: the ripple is gone and only the 10 kHz IF remains, with the 5 kHz envelope (two IF cycles per envelope cycle). - DC levels are as designed: I is centered at 1.2 V (range about 0.6 to 1.75 V) and Q around 1.0 V (range about 0.6 to 1.6 V) (*plot*), so all signals stay inside the 0 to 3.3 V supply range. - I and Q have different shapes because they are the sine and cosine components of the same IF, which is what the second mixer needs.

Stage 4: Recovered message compared with the original



Message vs recovered

What it shows - Top: the original message V(vm), a 5 kHz sine of about ±0.82 V. Bottom: the recovered signal V(vmre), also a clean 5 kHz sine of about ±0.83 V centered on 0 V (*plot*). - Amplitude and frequency match, so the carrier, the IF and the DC offsets are all removed correctly. - The recovered wave lags the original by about 10 µs, which matches the filter group delay (≈ $2/\omega_c$ = 10.6 µs). - The spike near t = 0 (up to about 2.8 V) is the start-up transient of the filters and settles within about 0.1 ms.

Summary

Stage	Key result	Source
RF input	AM, µ = 0.5, envelope 0.825 V to 2.475 V	plot
Op-amp	DC gain 79.7 dB, UGF about 206 MHz (Cc = 0.1 pF), 405 µA	ngspice
Low-pass filter	-3 dB at 29.9 kHz, -48 dB at 190 kHz, 60 dB/decade	ngspice
I/Q after filter	10 kHz IF with 5 kHz envelope, 190 kHz ripple removed	plot
Recovered message	about ±0.83 V, 5 kHz, delay about 10 µs	plot